Given Data **exp 7**

Bus Data

Bus	Type	PG	QG	PD	QD	V
1	Slack	0	0	0	0	1.06
2	PV	0	0	270	130	1.00
3	PQ	230	0	0	0	1.04

Line Data

From	To	R	X	B/2	Tap
1	2	0.03	0.04	0	1
2	3	0.011	0.015	0	1
1	3	0.024	0.035	0	1

Base MVA = 100

Step 1: Calculate Line Admittances

Between Bus 1–2

$$Z_{12} = 0.03 + j0.04$$
$$Y_{12} = \frac{1}{0.03 + j0.04}$$

Multiply conjugate:

$$Y_{12} = \frac{0.03 - j0.04}{0.03^2 + 0.04^2}$$
$$Y_{12} = 12 - j16$$

Between Bus 2–3

$$Z_{23} = 0.011 + j0.015$$
$$Y_{23} = \frac{1}{0.011 + j0.015}$$
$$Y_{23} = 30.556 - j41.667$$

Between Bus 1–3

$$Z_{13} = 0.024 + j0.035$$
$$Y_{13} = 13.841 - j20.185$$

Step 2: Formation of Ybus Matrix

Diagonal Elements:

$$\begin{aligned} Y_{11} &= Y_{12} + Y_{13} \\ &= 25.841 - j36.185 \\ Y_{22} &= Y_{12} + Y_{23} \\ &= 42.556 - j57.667 \\ Y_{33} &= Y_{13} + Y_{23} \\ &= 44.397 - j61.852 \end{aligned}$$

Off Diagonal:

$$Y_{12} = -12 + j16$$
$$Y_{13} = -13.841 + j20.185$$
$$Y_{23} = -30.556 + j41.667$$

Therefore,

$$Y_{bus} = \begin{bmatrix} 25.841 - j36.185 & -12 + j16 & -13.841 + j20.185 \\ -12 + j16 & 42.556 - j57.667 & -30.556 + j41.667 \\ -13.841 + j20.185 & -30.556 + j41.667 & 44.397 - j61.852 \end{bmatrix}$$

Step 3: Calculate Scheduled Powers

Real Power:

$$P = \frac{PG - PD}{100}$$

Bus 1:

$$P_1 = 0$$

Bus 2:

$$P_2 = \frac{0 - 270}{100}$$
$$P_2 = -2.7$$

Bus 3:

$$P_3 = \frac{230 - 0}{100}$$
$$P_3 = 2.3$$

Therefore,

$$P = \begin{bmatrix} 0 \\ -2.7 \\ 2.3 \end{bmatrix}$$

Reactive Power:

$$Q = \frac{QG - QD}{100}$$
$$Q = \begin{bmatrix} 0 \\ -1.3 \\ 0 \end{bmatrix}$$

Step 4: Initial Conditions

$$V = \begin{bmatrix} 1.06 \\ 1.00 \\ 1.04 \end{bmatrix}$$
$$\delta = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 5: Fast Decoupled Iteration

Using:

$$\Delta P = B' \Delta \delta$$

and

$$\Delta Q = B'' \Delta V$$

After solving mismatch equations and updating voltage values iteratively:

Iteration 1 → Voltage corrected

Final Results

Bus	Voltage (pu)	Angle (deg)
1	1.0600	0
2	1.0097	−0.11415
3	1.0400	1.2566

Iteration 2 → Angle corrected

Iteration 3 → Convergence achieved

Final Results

Bus	Voltage (pu)	Angle (deg)
1	1.0600	0
2	1.0097	−0.11415
3	1.0400	1.2566